

Question

Write a Java program to demonstrate multilevel inheritance and perform addition and subtraction using inherited methods.

Steps for Execution

1. Create a class FirstA with:
 - display() method to print "hello".
 - sum() method to add two integers.
2. Create a class SecondA that extends FirstA.
 - Add display2() method.
 - Add sub() method to subtract two integers.
3. Create a class ThirdA that extends SecondA.
 - Add display3() method.
4. Create the MultilInheritance class containing the main() method.
5. Create an object t of ThirdA.
6. Call display() using the ThirdA object. This method is inherited from FirstA.
7. Call sub(2,7) and display the result.
8. Call sum(2,7) and display the result.
9. Compile and execute the program.

Algorithm

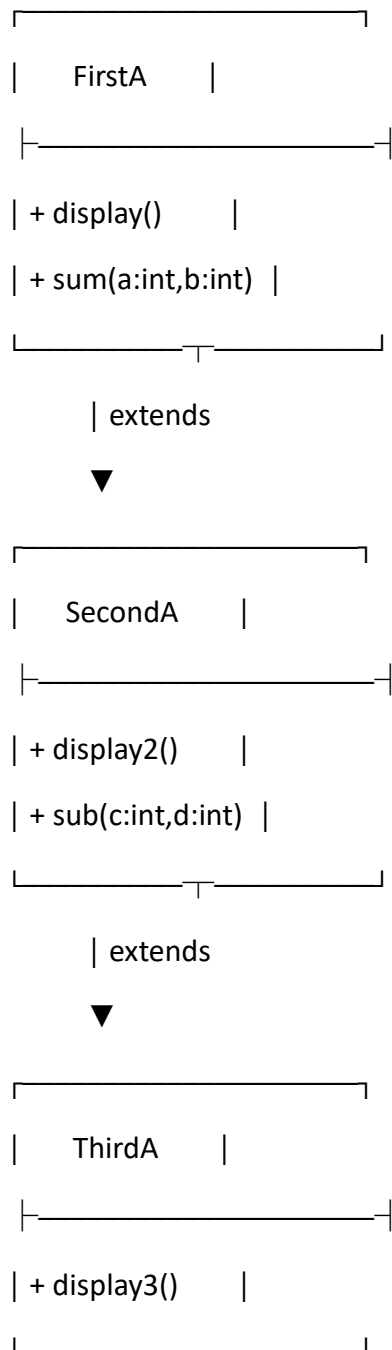
1. **Start**
2. Define class FirstA.
3. Define display() to print "hello".
4. Define sum(a,b) to return a+b.
5. Define class SecondA extending FirstA.
6. Define display2() and sub(c,d) in SecondA.
7. Define class ThirdA extending SecondA.
8. In the main() method, create an object of ThirdA.
9. Call the inherited display() method.

10. Calculate subtraction using sub(2,7) and display the result.

11. Calculate addition using sum(2,7) and display the result.

12. **Stop**

UML Class Diagram



This is **multilevel inheritance** because ThirdA inherits from SecondA, and SecondA inherits from FirstA.

Code

```
package javaprograms;
```

```
class FirstA {  
    void display() {  
        System.out.println("hello");  
    }  
  
    int sum(int a, int b) {  
        return a + b;  
    }  
}
```

```
class SecondA extends FirstA {  
    void display2() {  
        System.out.println("java program");  
    }  
  
    int sub(int c, int d) {  
        return c - d;  
    }  
}
```

```
class ThirdA extends SecondA {  
    void display3() {  
        System.out.println("arrays");  
    }  
}
```

```
class ThirdA extends SecondA {  
    void display3() {  
        System.out.println("arrays");  
    }  
}
```

```
class ThirdA extends SecondA {  
    void display3() {  
        System.out.println("arrays");  
    }  
}
```

```
public class MultiInheritance {  
    public static void main(String[] args) {  
        ThirdA t = new ThirdA();  
  
        t.display();  
  
        int res = t.sub(2, 7);  
        System.out.println("res is:" + res);  
  
        int result = t.sum(2, 7);  
        System.out.println("res is : " + result);  
    }  
}
```

Output

hello

res is:-5

res is :9